

Mohammad Amir Jamali

Senior Software & Platform Architect | AI, Embeddings & Network Systems

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Location: Open to Relocation (EU / Worldwide)

PROFESSIONAL SUMMARY

Senior Software Engineer and Platform Architect with 8+ years of experience engineering high-throughput backend services, distributed cloud architectures, and AI-enabled data pipelines. Proven expertise in scalable microservices, Vector Embeddings & RAG systems, Linux networking/infrastructure, and low-level embedded system design. Strong background in Python, TypeScript, C/C++, Docker/Kubernetes, and performance-critical optimizations for international enterprises.

TECHNICAL SKILLS

Languages:	Python, TypeScript, JavaScript, Java, C/C++, SQL, Bash/Shell Scripting
AI, Embeddings & ML:	Vector Embeddings, Semantic Search, RAG, LLM Integration, PyTorch, TensorFlow
Networking & Infrastructure:	TCP/IP, DNS, VPN, Firewalls, Linux Server Administration (Ubuntu/CentOS), Nginx
Cloud & DevOps:	Docker, Kubernetes, Apache Kafka, CI/CD, Jenkins, GitHub Actions
Backend & Frameworks:	FastAPI, Django, Nest.js, Node.js, REST APIs, Microservices Architecture
Databases & Storage:	PostgreSQL, MySQL, Redis, Elasticsearch (Vector Search), MongoDB

EMBEDDED SYSTEMS & MICROCONTROLLERS

Microprocessor & Assembly (EMU8086, AVR):

- Microprocessor structure and 8086/Pentium architecture
- Assembly programming with EMU8086 & 8086 CPU design
- USART registers & modules in AVR microcontrollers

ARM Microcontroller Training (Keil, LPC1768):

- ARM microcontroller families & LPC1768 initialization
- GPIO, NVIC, ADC, Timer, PWM, DAC, USART, RTC modules

WORK EXPERIENCE

Senior Software & Systems Engineer

Pangian — San Francisco, CA, USA (Remote)

Aug 2023 – Feb 2026

Silicon Valley

- Architected and scaled distributed backend microservices using **TypeScript** and **NestJS** handling high-concurrency workflows.
- Designed high-performance **Vector Embedding pipelines and semantic search modules** utilizing LLMs and Elasticsearch/Redis.
- Administered cloud network environments, ensuring robust security, low-latency routing, and automated deployments via **Docker** and **Kubernetes**.

Senior Software Engineer

Enviroble — London, UK (Remote)

Mar 2022 – Apr 2023

Supported by UCL BaseKX

- Engineered scalable backend microservices translating complex environmental datasets into responsive APIs.
- Applied statistical models and machine learning techniques for predictive analytics and data ingestion.

Software Engineer / Co-Founder

Tradeoption — Mashhad, Iran

Jan 2020 – Feb 2022

Financial Platform

- Developed low-latency algorithmic trading architecture, real-time market data ingestion, and secure WebSocket communications.

Software Developer / Co-Founder

Class724 — Birjand, Iran

May 2015 – Nov 2019

Science & Technology Park of South Khorasan

- Co-founded and engineered an interactive Learning Management System (LMS) supporting thousands of concurrent users with Docker/Linux.

EDUCATION & CERTIFICATIONS

Master of Information Technology Engineering

Islamic Azad University

2017 – 2019

* Anabin Verified: Institution **H+** | Equivalence **Class A5**

Bachelor of Science in Industrial Engineering

Payame Noor University, Birjand Branch

2009 – 2014

* Anabin Verified: Institution **H+** | Equivalence **Class A4**

Certifications & Professional Development: Microsoft Certified IT Professional (MCITP)

PUBLICATIONS & RESEARCH

Predicting Bitcoin Price Using Bayesian Neural Networks (Master Thesis, 2019) | **An Introduction to Quantum Computing** (2021) | **Comparison of LG Trader and EMA** (2019) | **Secure & Trustworthy Data Management for Explainable ML** (Working Paper)